PharmAchieve

CHRONIC OBSTRUCTIVE PULMONARY DISEASE (COPD)

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION…

Providing Care: Clinical Care

Providing Care: Distribution

Knowledge & Expertise

Communication & Collaboration

Leadership & Stewardship

Professionalism

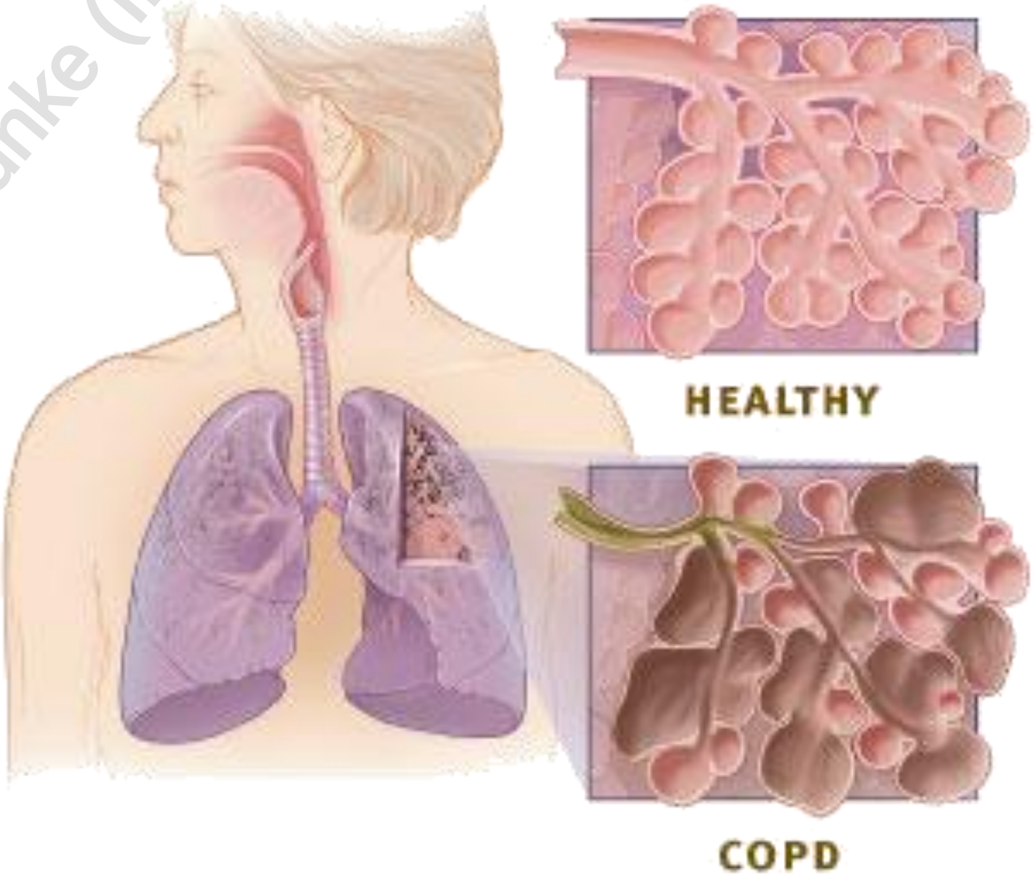
LEGEND

AATD	Alpha-1 Antitrypsin Deficiency
ADR	Adverse Drug Reaction
ADL	Activities of Daily Living
AECOPD	Acute Exacerbation of COPD
AF	Atrial Fibrillation
CAAT	Chronic Airway Assessment Test
CHF	Congestive Heart Failure
COPD	Chronic Obstructive Pulmonary Disease
CTS	Canadian Thoracic Society
EOS	Eosinophils
FEV1	Forced Expiratory Volume In 1 Second
FVC	Forced Vital Capacity
GOLD	Global Initiative for Chronic Obstructive Lung Disease
ICS	Inhaled Corticosteroid
LABA	Long-Acting Beta-2 Agonist
HIV	Human Immunodeficiency Virus
LAMA	Long-Acting Muscarinic Antagonist
LRTI	Lower Respiratory Tract Infection
MAC	Mycobacterium Avium Complex
mMRC	Modified Medical Research Council
Mod-Sev	Moderate-severe
NAC	N-acetyl Cysteine
NACI	National Advisory Committee On Immunizations
NIV	Non-Invasive Ventilation
NRT	Nicotine Replacement Therapy
O2	Oxygen
PO	By Mouth
PRN	As Needed
PT	Patient
RSV	Respiratory Syncytial Virus
SABA	Short-Acting Beta-2 Agonist
SOB	Shortness Of Breath
TX	Therapy
Tdap	Tetanus Diphtheria Acellular Pertussis
VAS	Visual Analogue Scale
WBC	White Blood Cells

PATHOPHYSIOLOGY

- Progressive, partially reversible airflow limitation
- Inflammation throughout central and peripheral airways, lung parenchyma, and pulmonary vasculature resulting in mucus hypersecretion and/or lung changes
- Pathophysiological hallmark of COPD: expiratory flow limitation

"COPD is a heterogeneous lung condition characterized by chronic respiratory symptoms (dyspnea, cough, sputum production and/or exacerbations) due to abnormalities of the airways (bronchitis, bronchiolitis) and/or alveoli (emphysema) that cause persistent, often progressive, airflow obstruction." – *GOLD 2026*



CONTRIBUTORS

Risk factors

- Smoking (leading cause)
- Air pollution/biomass cooking and heating
- Long-term dust & chemical exposure (e.g. occupation-related)
- Genetics (AATD deficiency)
- Age ≥ 35 years old
- Childhood infections
- Tuberculosis-associated COPD
- HIV-associated COPD
- Premature birth weight and low birth weight
- Recurrent LRTIs

Comorbidities

- Cardiovascular disease
- Metabolic syndrome
- Skeletal muscle dysfunction
- Gastroesophageal reflux disease
- Renal insufficiency
- Sleep apnea
- Depression
- Anxiety
- Osteoporosis
- Lung cancer

Drug Factors

- Beta-blockers
- ACE inhibitors (causes persistent dry cough, unrelated to COPD)
- Note: not causative, may exacerbate or mimic disease or limit response to therapy

CLINICAL PRESENTATION

Signs

- Reductions in FEV1/FVC
- Hypoxia with or without hypercapnia
- Physiologic responses to chronic hypoxia (polycythemia, pulmonary hypertension, cor pulmonale)

Symptoms

- Dyspnea
- Exercise limitations
- Chronic cough with or without sputum
- Recurrent wheezing
- Acute respiratory events (e.g. exacerbations)

Diagnosis

Symptom History + Risk Factors + Spirometry

- Spirometry Diagnostic Criteria:
  - ❑ $FEV_1/FVC < 0.7$ after bronchodilator use
- GOLD Spirometry Severity Criteria:
  1) Gold 1 - Mild: $FEV_1 \geq 80\%$ predicted
  2) Gold 2 - Moderate: $50\% \leq FEV1 < 80\%$ predicted
  3) Gold 3 - Severe: $30\% \leq FEV1 < 50\%$ predicted
  4) Gold 4 - Very Severe: $FEV1 < 30\%$ predicted

COMPARING ASTHMA & COPD

	Asthma	COPD
Pathology	A combination of genetic and environmental causes, leading to increased contractability of the respiratory smooth muscles. Pulmonary constriction is associated with triggers such as cold air, allergens, overexertion, and inhaled irritants.	COPD is an accumulation of damage/scar tissue from prolonged exposure to inhaled chemicals (e.g. long-term smoking or occupational exposures). This leads to reduced elasticity of the lungs due to scarring (emphysema) and hypersecretion of phlegm (chronic bronchitis).
Onset	Usually under 40 years old	Usually ≥ 35 years old
Smoking history	No correlation	Usually 10+ years of smoking
Sputum	Rarely	Frequently
Allergies	Frequently, especially if there is a family history	Infrequent
Course	Stable with exacerbations	Progressively worsening with increasing frequency & severity of exacerbations
Clinical symptoms	Shortness of breath, chest tightness, wheezing and coughing attacks worsened by respiratory viruses. Intermittent in nature and variable based on trigger exposure.	Shortness of breath, fatigue, productive cough, chest tightness, and weight loss/ muscle loss (inability to stay active). Persistent symptoms and worsens with increased levels of exertion.

CLASSIFICATION OF SEVERITY

mMRC Dyspnea Scale

- Grade 0: SOB with strenuous exercise
- Grade 1: SOB when walking up hill or when walking in a hurry
- Grade 2: Walking slower than peers of same age or stopping for breath when walking at usual pace
- Grade 3: Stopping for breath after walking 100 meters or after walking for a few minutes
- Grade 4: SOB while performing ADLs

Chronic Airway Assessment Test (CAAT)

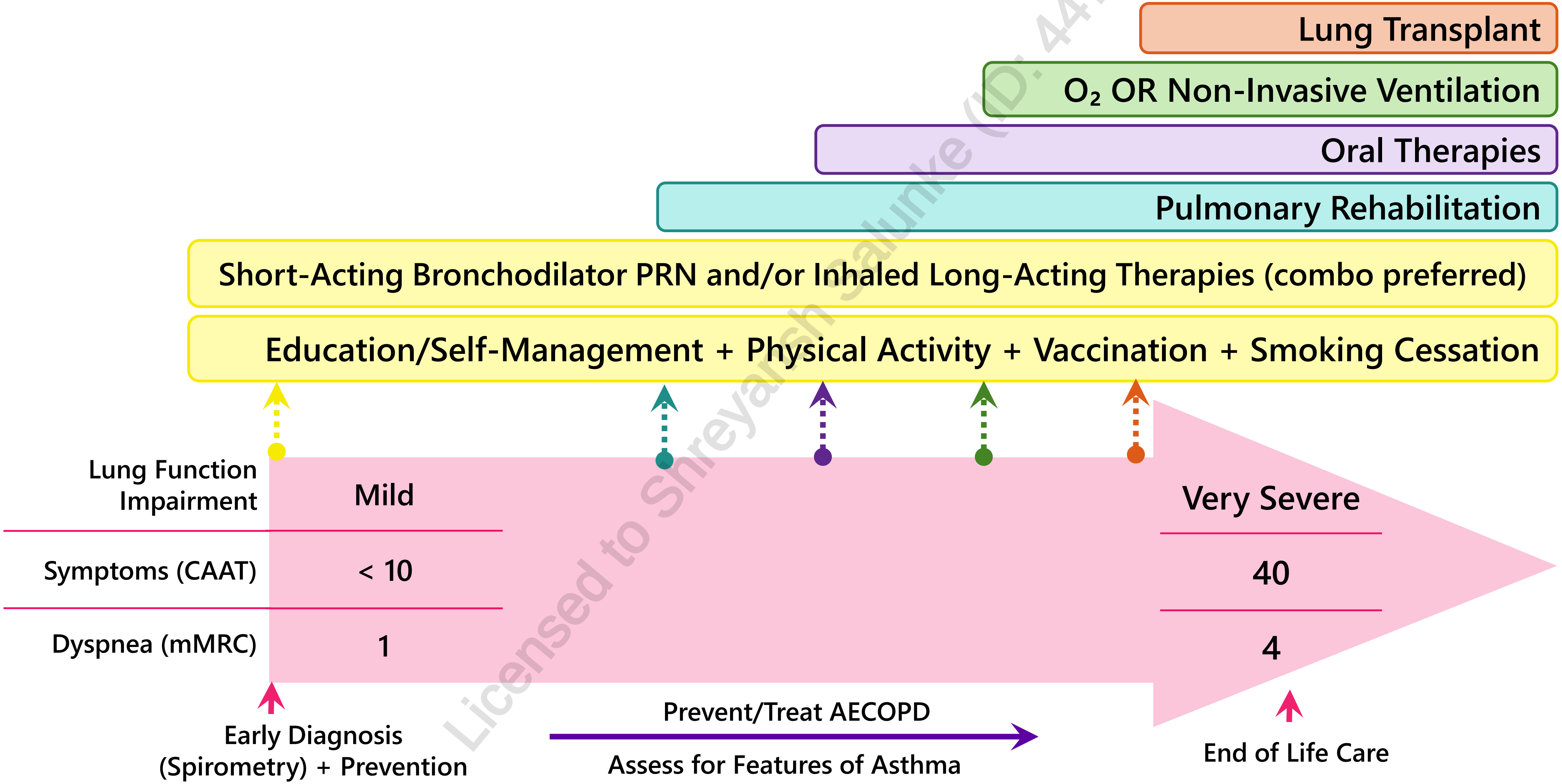
- 8-question assessment of common COPD symptoms and side effects
- 5-point scale for each question
- Total score ranges from 0 to 40
- Higher scores indicate greater impact on health

E		≥1 moderate or severe exacerbations in past year	Exacerbations
A	B	0 moderate or severe exacerbations in past year	
mMRC 0 - 1 OR CAAT < 10	mMRC ≥ 2 OR CAAT ≥ 10		
Symptoms			

GOALS OF THERAPY

- ✓ Prevent disease progression by providing smoking cessation support
- ✓ Reduce frequency and severity of exacerbations
- ✓ Alleviate dyspnea
- ✓ Improve exercise tolerance and daily activity → pulmonary rehabilitation
- ✓ Treat exacerbations and complications of the disease
- ✓ Improve health status and concomitant disease burden
- ✓ Reduce mortality related to cardiovascular complications of COPD (AF, CHF, pulmonary hypertension, cor pulmonale)

COPD MANAGEMENT: OVERVIEW



COPD MANAGEMENT – NON-PHARMACOLOGICAL OPTIONS

Vaccinations Per NACI Recommendations	• Influenza • Pneumococcal • SARS-CoV-2 • Tdap • RSV • Herpes zoster (shingles)
Smoking Cessation	• Counselling should be offered at every visit • Nicotine Replacement Therapy (NRT), bupropion, varenicline
Symptom Self Management	• Pulmonary rehabilitation (e.g. exercise training + disease-specific education) • May be delivered virtually
Home oxygen	• For severe hypoxia at rest, improves survival • Current smoking is a contraindication

INITIAL COPD MANAGEMENT: PHARMACOLOGICAL

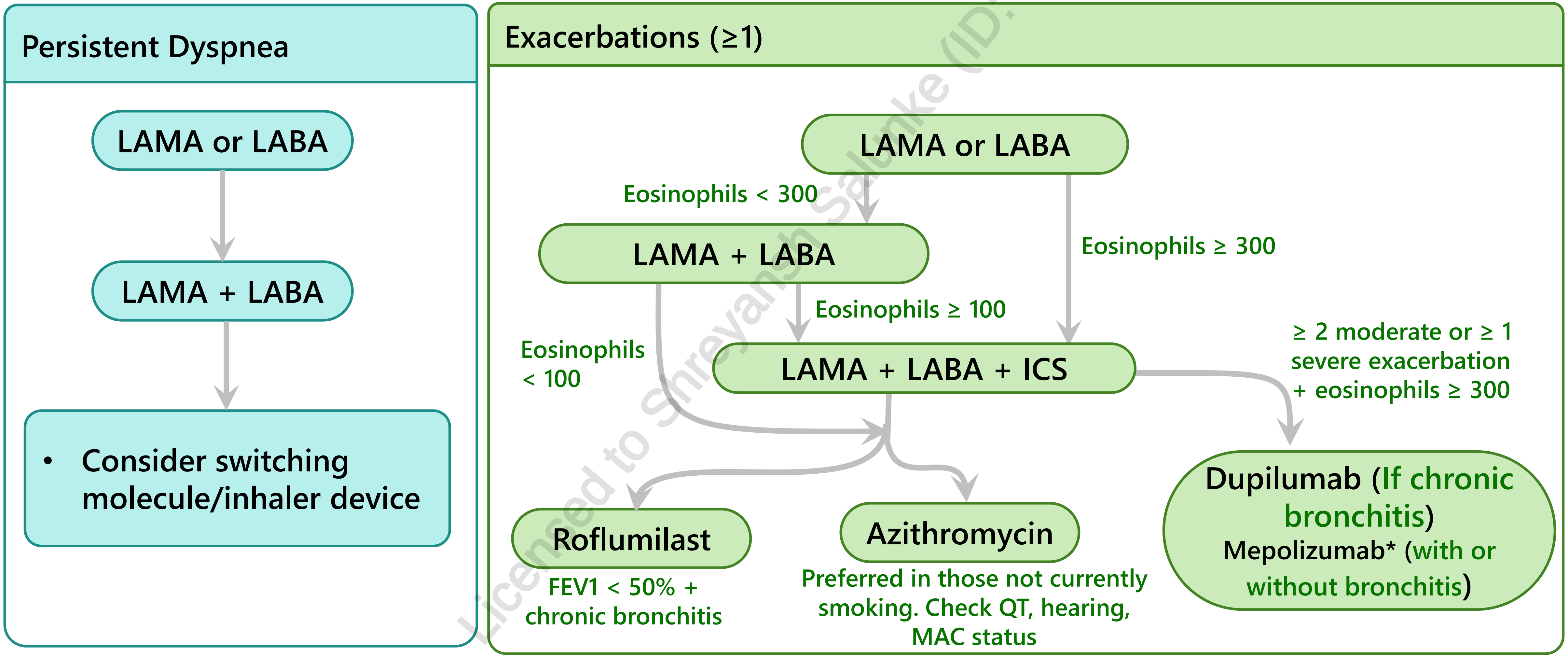
		Exacerbations
E LAMA + LABA* (consider LAMA + LABA + ICS* if serum eos ≥ 300)		≥1 moderate or severe exacerbations in past year
A SABA or SAMA or LABA or LAMA	**B** LAMA + LABA*	0 moderate or severe exacerbations in past year
mMRC 0 - 1 OR CAAT < 10	mMRC ≥ 2 OR CAAT ≥ 10	
Symptoms		

- Short-acting bronchodilator PRN can be used at any stage for dyspnea or wheezing

* Single inhaler therapy may be more convenient and effective than multiple inhalers

TAILORING COPD MANAGEMENT

WHAT'S NOT OPTIMALLY CONTROLLED?



* Under regulatory review but not officially approved yet for COPD

WHEN TO USE AN INHALED CORTICOSTEROID (ICS)

Strongly favours use

- History of hospitalization(s) for COPD exacerbations despite being on appropriate long-acting bronchodilator maintenance therapy
- ≥ 2 moderate COPD exacerbations in the last year despite being on appropriate long-acting bronchodilator maintenance therapy
- Blood eosinophils ≥300 cells/μL
- History of or concurrent asthma

Favours use

- 1 moderate COPD exacerbation/year despite being on appropriate long-acting maintenance therapy
- Blood eosinophils 100 - < 300 cells/μL

Against use

- Repeated pneumonia
- Blood eosinophils < 100 cells/μL
- History of mycobacterial infection

Note: Eosinophil values are approximations; they should be interpreted as a continuum

ACUTE EXACERBATIONS (AECOPD)

- Acute worsening of dyspnea, sputum purulence and/or sputum production in the span of a few days (up to 14 days) leading to an increased use of maintenance medications and/or addition of other medications
- Most frequently caused by infections (bacterial, viral) or environmental pollutants or other conditions (e.g. pulmonary embolism, acute heart failure)
- AECOPD leads to increased medical visits, hospitalization, reduced lung function and mortality
- Check inhaler technique, precipitants, smoking status

AECOPD CLASSIFICATION & TREATMENT

Severity is determined based on several factors, e.g. dyspnea (VAS score), respiratory rate, heart rate, Oxygen Saturation (SaO2), C-Reactive Protein (CRP), and arterial blood gases

Severity	Treatment
Mild	• SABA +/- SAMA • +/- Antibiotics*
Moderate	• SABA +/- SAMA • +/- Antibiotics* • +/- Short course PO corticosteroids**
Severe	• Respiratory support/oxygen • SABA +/- SAMA • Antibiotics* • Corticosteroids (IV or PO)**

*Antibiotics are recommended for patients during an exacerbation with any of the following:

- Prior positive sputum bacteria culture
- Requiring mechanical ventilation
- At least 2 of the following: increased dyspnea, sputum volume, sputum purulence or fever – 1 of the 2 *must* be increased sputum purulence

**PO corticosteroid: dose of 40 mg prednisone-equivalent for 5 days

ANTIBIOTIC TREATMENT RECOMMENDATIONS FOR AECOPD

- Antibiotic selection should be based on local bacterial resistance patterns
- Recommended length of therapy is 5 – 7 days (≤ 5 days for outpatient treatment)
- To minimize resistance, switch antibiotic class if therapy is repeated within 3 months

COPD Group	Symptoms & Risk Factors	Probable Pathogens	First-line therapy
• Simple (COPD without risk factors)	• Smokers • FEV1 > 50% • ≤ 3 exacerbations/ year	• *Streptococcus pneumoniae* • *Haemophilus influenzae* • *Moraxella catarrhalis*	• Amoxicillin • Doxycycline • Azithromycin • Clarithromycin • Sulfamethoxazole and trimethoprim
• Complicated (COPD with risk factors)	Simple + at least one: • FEV1 < 50% predicted • ≥ 4 exacerbations/ year • Ischemic heart disease • Use of home oxygen • Chronic PO corticosteroids	Same as simple plus: • *Klebsiella pneumoniae* • *Pseudomonas aeruginosa* • Beta-lactam resistant pathogens	In order of preference: • Levofloxacin, Ciprofloxacin or moxifloxacin • Amoxicillin-clavulanate

MONITORING PARAMETERS

Telehealth (virtual, hybrid) care models may improve healthcare access, outcomes and affordability

Efficacy Endpoints		
Parameter	**Degree of Change**	**Timeframe**
Symptoms (e.g. exacerbations, dyspnea, exercise tolerance, cough, sputum)	Minimal	At each visit
Lung function (FEV1%)	Minimal	Regular intervals (e.g. yearly)
Antibiotics	Reduction in sputum volume, purulence, dyspnea and fever	Within 24 – 48 hours

Safety Endpoints		
Parameter	**Degree of Change**	**Timeframe**
Inhaler adherence and appropriate use	None	Address and manage at each visit
Medication-induced adverse effects	Prevent/ minimize	
Comorbidities (e.g. smoking, cardiovascular disease)	Eliminate/ control	

TAKEAWAYS

- COPD is a chronic lung condition characterized by dyspnea, cough, sputum production and/ or exacerbations
- Spirometry of $FEV_1/FVC < 0.7$ after bronchodilator use confirms the diagnosis
- Initial treatment is based on severity during diagnosis
  - Group A: No moderate/severe exacerbation in past year, mMRC 0-1, CAT <10
    - Treatment: SABA, SAMA, LABA or LAMA
  - Group B: No moderate/severe exacerbation in past year, mMRC >1, CAT ≥ 10
    - Treatment: LABA + LAMA
  - Group E: Any moderate or severe exacerbation in past year
    - Treatment: LABA + LAMA (+ ICS if eosinophils ≥300)
- Adjunctive therapy for exacerbations (if eligible) may include azithromycin, roflumilast or dupilumab

- AECOPD: SABA +/- SAMA, +/- PO prednisone, +/- antibiotics
  - Systemic corticosteroids are recommended for up to 5 days in moderate/severe exacerbations
  - Antibiotics are only indicated for prior positive sputum bacteria culture, mechanical ventilation or increased sputum purulence + at least 1 of the following: increased dyspnea, sputum volume, or fever

CASE 1

JW is a 67-year-old male who presents to your outpatient clinic for his routine appointment. He reports that his shortness of breath has worsened over the past several weeks and is not relieved by his salbutamol inhaler. During the visit, his modified Medical Research Council (mMRC) dyspnea scale score is 1, and his Chronic Airway Assessment Test (CAAT) score is 8. JW's medical history includes chronic obstructive pulmonary disease (COPD) (diagnosed five years ago), depression (diagnosed ten years ago), and hypertension (diagnosed eleven years ago). He mentions that he takes all his medications as prescribed, which include salbutamol 100 mcg 1-2 puffs q4h as needed, sertraline 50 mg PO daily, and ramipril 15 mg PO daily. Additionally, he takes a multivitamin and vitamin C daily for general health. JW has a social history that reveals he smoked one pack of cigarettes a day for 25 years but quit five years ago. He adheres to a diet that aligns with Canada's Food Guide and does not consume alcohol. During your assessment of JW's inhaler technique, you find that he is using it correctly.

1. What is the best course of action at this time?
   a) Discontinue salbutamol and start budesonide/formoterol prn
   b) Start oral prednisone for 5 days
   c) Start a umeclidinium/vilanterol combination inhaler as maintenance therapy and continue salbutamol prn
   d) Start formoterol or tiotropium inhaler as maintenance therapy and continue salbutamol prn

CASE 1

Four weeks later, JW returns to your clinic for a follow-up visit. He reports that he has been adherent to his Spiriva® Respimat (tiotropium bromide monohydrate) 2.5 mcg inhaler, taking 2 puffs daily; however, he still requires his salbutamol inhaler 100 mcg, using 1-2 puffs 3 to 4 times per day. JW reports that 2 weeks ago, he experienced fever, increased thick mucus, and worsening shortness of breath. After these symptoms arose, he consulted his family doctor, who diagnosed him with an acute exacerbation. JW has since completed a 5-day course of oral prednisone and antibiotics. Since the exacerbation, JW notes that he walks much more slowly than his wife and often has to pause to catch his breath during leisurely walks around his neighbourhood.

2. What would be the best approach to control JW's worsening symptoms?
   a) Continue the Spiriva Respimat® and salbutamol. Refer JW for a pulmonary function test to assess for concurrent asthma
   b) Add formoterol (Oxeze® Turbuhaler) to JW's current regimen
   c) Switch the tiotropium (Spiriva Respimat®) to tiotropium/olodaterol (Inspiolto® Respimat) 2.5/2.5 mcg per actuation, 2 puffs once daily
   d) Switch tiotropium (Spiriva Respimat®) to umeclidinium/vilanterol (Anoro® Ellipta) 62.5/25mcg per actuation, 1 puff once daily

CASE 1

3. Which of the following is an appropriate counselling point for the inhaler selected in Question 2?
   a) It is essential to keep track of the number of doses used, as the selected device has no dose indicator to let you know when the cartridge is empty
   b) Patient should inhale sharply and forcefully so that the powder is released from the device into the lungs
   c) The patient should rinse his mouth after every use to prevent oral thrush from developing
   d) The patient may experience include dry mouth, urinary retention and blurred vision

4. What should you counsel the patient regarding his salbutamol use?

   a) JW should keep using salbutamol as reliever therapy
   b) JW should be switched to budesonide/formoterol as reliever therapy
   c) JW should discontinue salbutamol and initiate ipratropium as reliever therapy
   d) JW should also add budesonide/formoterol as reliever therapy

CASE 2

MJ, a 76-year-old male, presents to your emergency department with increasing dyspnea and sputum purulence over the last 3 days. He reports consistent fevers, lethargy and increased sputum as his chief complaint. His objective findings include an O2 saturation of 89%, a respiratory rate of 29 breaths/min, and a temperature of 37.8 °C. His chest X-ray reveals mild bibasilar atelectasis and hyperinflation. His WBC is 11 × 10^9 cell/L. JW's past medical history includes COPD (diagnosed 10 years ago), with one mild acute exacerbation five years ago (managed at home using a short course of antibiotics and oral prednisone), anxiety (diagnosed 5 years ago), and dyslipidemia (diagnosed 11 years ago). His medication history includes salbutamol 100 mcg 1-2 puffs q4h PRN, tiotropium/olodaterol 2.5/2.5 mcg 2 puffs daily, sertraline 50 mg po daily and atorvastatin 20 mg po daily. He has a social history of smoking for 30 years, at a rate of 1 pack a day, but quit 16 years ago.

1. From which of the following COPD-related conditions is JW likely suffering?

    a) Pulmonary embolism
    b) Cor pulmonale
    c) COPD exacerbation
    d) Pneumonia

CASE 2

2. Which of the following is the BEST recommendation for acute management?

   a) Stop tiotropium/olodaterol, start O2, increase salbutamol to 4 puffs q4h prn and start tiotropium monotherapy and methylprednisolone
   b) Continue home medications, increase salbutamol to 4 puffs q4h standing+prn, start amoxicillin 500mg PO TID and start oral prednisone
   c) Continue home medications, start high-flow O2 therapy, start levofloxacin 750mg IV daily, increase salbutamol to 4 puffs q2h prn
   d) Continue home medications, start ipratoprium 2 puffs q2h prn, start IV hydrocortisone and epinephrine infusion

3. What is the MOST likely pathogen responsible for MJ's exacerbation

   a) *S. pneumoniae*
   b) *P. aeruginosa*
   c) *E. coli*
   d) *C. pneumoniae*

CASE 2

MJ has been hospitalized for the past 5 days and is now recovering. His O2 saturation is 95%, and his shortness of breath has improved significantly. The medical team is planning his discharge, with a referral pending for outpatient pulmonary rehabilitation. The physician has consulted with you regarding MJ's maintenance COPD therapy. You review MJ's most recent bloodwork and note that his eosinophil count is 280 cells/uL.

4. Which of the following would be the most appropriate next step in management?
   a) Switch tiotropium/olodaterol to fluticasone furoate/umeclidinium/vilanterol (Trelegy Ellipta®)
   b) Continue current regimen, as MJ's eosinophil count is < 300 cells/uL
   c) Continue current regimen, as MJ has only had one exacerbation in the past year and thus does not meet criteria for Class E
   d) Add azithromycin 500mg po three times weekly, assuming MJ's QT interval is normal and he is MAC negative

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CHANGE LOG

July 2025

- Formatting changes made throughout
- Updated legend
- Slide 9: Minor modification to diagram
- Slides 20-27: Case questions updated

October 2025

- Slide 16: New per 2025 GOLD update
- Slide 22: Updated takeaway

January 2026

- New guideline released and significant changes made throughout

March 2026

- Slide 2: Updated NAPRA competencies
- Slide 13: Added mepolizumab to algorithm + accompanying note
- Slide 16: Added note on determining the severity of AECOPD
- Slide 19: Updated takeaway for extra clarity